

storage, and Kubernetes clusters. This way, organisations can use GitOps and create modular AI.

Real world AI GitOps workflow

The real world AI GitOps workflow commences with the data scientist making changes to either the training code, the reference to the dataset, the logic used for the preprocessing, or the parameters of the model. The updated work is committed to a new feature branch, and a pull request is initiated. Before the new changes are merged into the main branch, an automated CI (continuous integration) pipeline has to check the code for correctness, its compliance with the data schema, and its validity with regard to the logical flow of the pipeline.

After the check is finished, the training pipeline loads the appropriate dataset version and executes the training process. Information such as the hyperparameters, the duration of training, the dataset version, and other metrics is collected through a tool for tracking experiments. The trained model is subsequently subjected to evaluation, and it has to meet certain benchmarks.

If the evaluation fails, it is deemed invalid and must be improved. Otherwise, the model is archived in a model registry and assigned a new version. The deployment repository is updated with the information regarding the new model, its image, and the necessary computational resources.

A GitOps controller like Argo CD or Flux finds out about the change and synchronises it with the Kubernetes environment. The new model may first see the light of day through a canary deployment, meaning that it will get only a small share of prediction requests. The monitoring solutions observe its performance in terms of the quality of predictions, latency and health of the system, along with the number of resources used. If the new model performs correctly, it is gradually promoted to be used in production.

If there is a major issue, the team can restore the previous working model by reverting the Git configuration.

Benefits, challenges, and best practices

GitOps is advantageous for the management of AI workloads in many ways. It ensures reproducibility by placing model configurations, pipeline specifications and infrastructure configurations under version control, allowing easy tracking of every change in Git. Thanks to Git, it is easy to find out who has made a specific change

and easily revert to a previous stable version if needed. Through automation, error during manual deployments is minimised, and all environments such as development, testing and production remain consistent.

On the other hand, there are many challenges associated with the implementation of GitOps for AI workloads. AI systems require the storage of huge datasets and model files which cannot be handled efficiently in regular Git repositories. For this reason, companies need additional tools (DVC, artifact storage, model registries, etc).

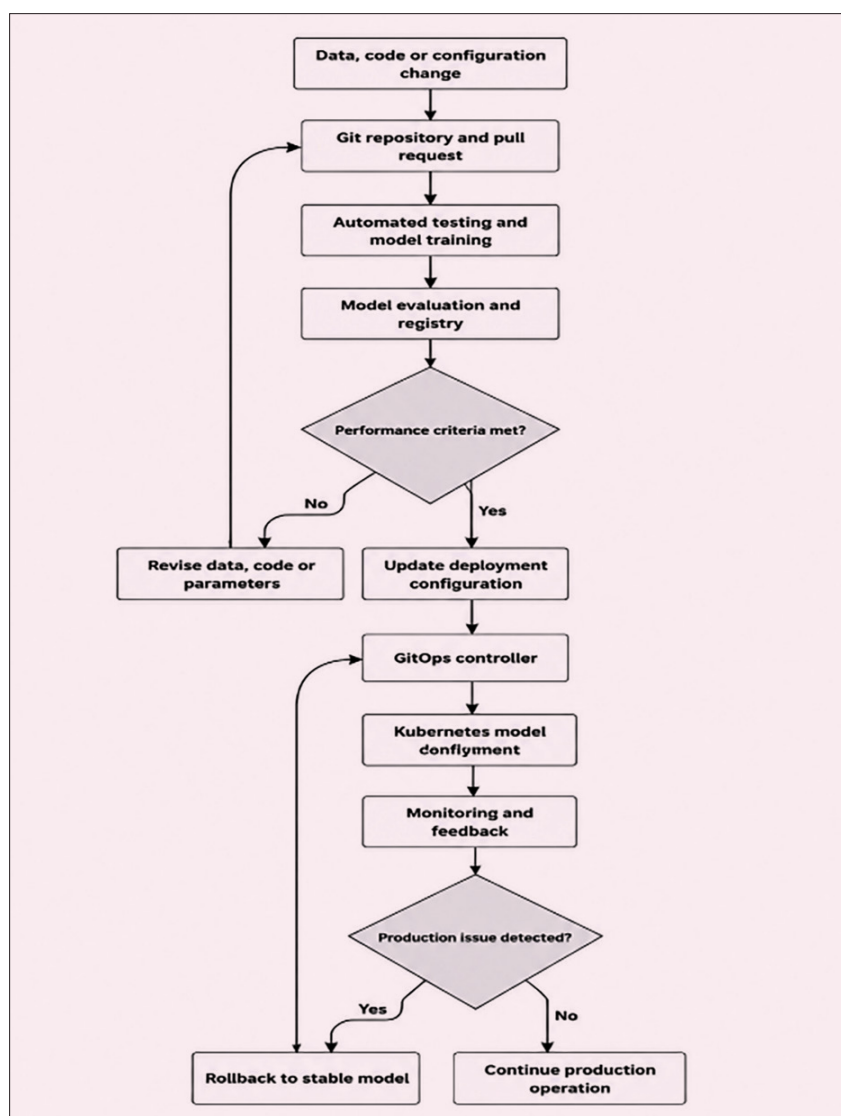


Figure 1: End-to-end GitOps workflow for AI workloads